

Exam 1

ECE 2201: Circuit Analysis I

Spring 2025

February 15, 2025

1. This exam is closed book, closed notes and a crib sheet has been provided to you with the exam. Print your name below.
2. Show all work on these pages. Show all work necessary to complete the problem. A solution without the appropriate work shown will receive no credit. A solution which is not given in a reasonable order will lose credit.
3. Show all units in solutions, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 75 minutes to work on this exam.

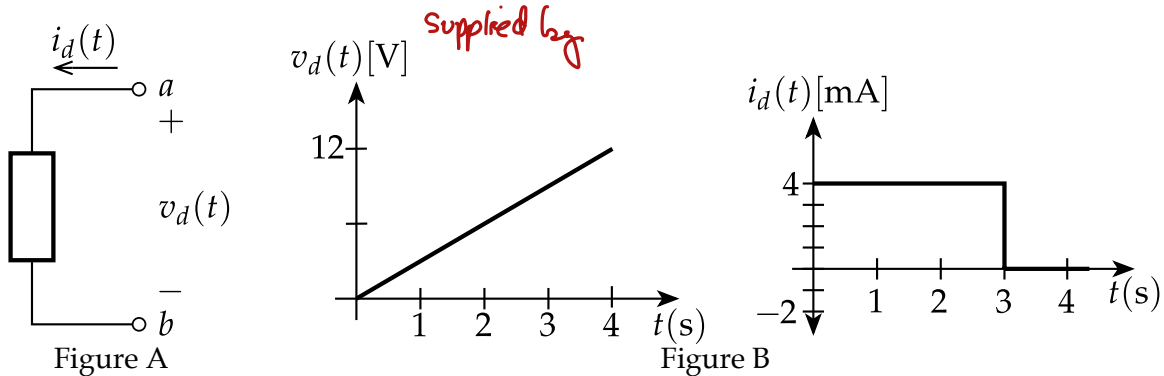
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Problem 1

A Device shown in Figure A has a current, $i_d(t)$, and a voltage, $v_d(t)$, as shown in Figure B.

- Write the piece-wise functions for the current, $i_d(t)$, and a voltage, $v_d(t)$.
- How much charge has moved into the device for $t > 0$.
- How much energy was delivered by the device for time $t > 0$.



$$(a) \quad v_d(t) = \begin{cases} 0 & t < 0 \\ 3\left[\frac{V}{s}\right] \cdot t & t > 0 \end{cases} \quad i_d(t) = \begin{cases} 0 & t < 0 \\ 4\text{[mA]} & 0 \leq t < 3\text{[s]} \\ 0 & t > 3\text{[s]} \end{cases}$$

b) Current flows into device: positive current $\rightarrow$ charge entering

$$i = \frac{dq}{dt} \Rightarrow dq = i dt \Rightarrow q = \int i dt$$

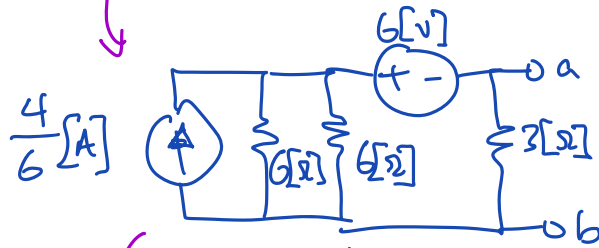
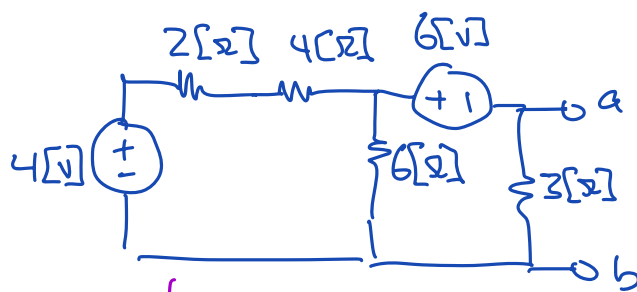
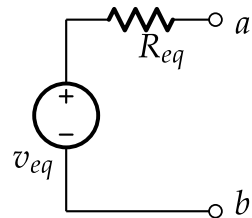
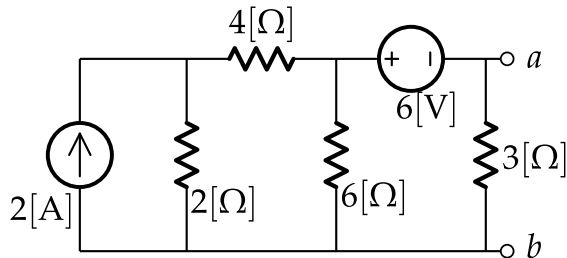
$$q = \int_0^{3\text{[s]}} 4\text{[mA]} dt = 4\left[\frac{\text{mC}}{\text{s}}\right] \cdot t \Big|_0^{3\text{[s]}} = 12\text{mC}$$

$$(c) \quad p = i \cdot v = \frac{dw}{dt} \Rightarrow dw = i \cdot v \cdot dt \Rightarrow w = \int i \cdot v \cdot dt$$

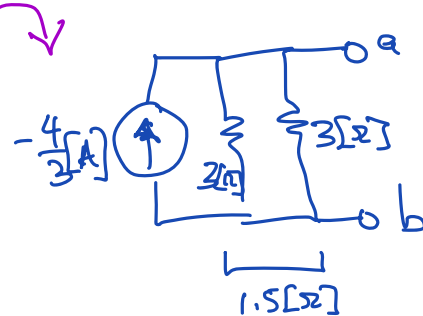
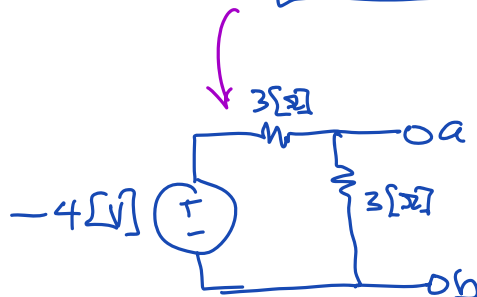
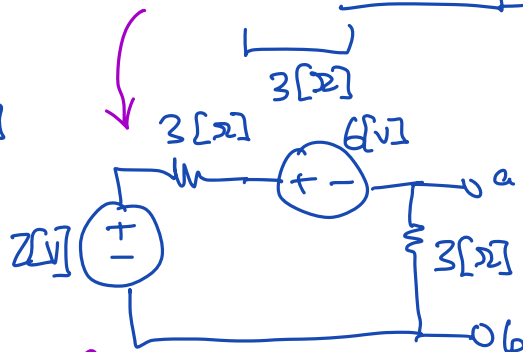
$$\begin{aligned} w &= - \int_0^{3\text{[s]}} 4\text{[mA]} \cdot 3\left[\frac{V}{s}\right] \cdot t dt = - \int_0^{3\text{[s]}} 12\left[\frac{\text{mW}}{\text{s}}\right] t dt = - \frac{12}{2}\left[\frac{\text{mW}}{\text{s}}\right] t^2 \Big|_0^{3\text{[s]}} \\ &= -6\left[\frac{\text{mJ}}{\text{s}^2}\right] t^2 \Big|_0^{3\text{[s]}} = -6 \cdot 9\text{[mJ]} = -54\text{[mJ]} \end{aligned}$$

Problem 2

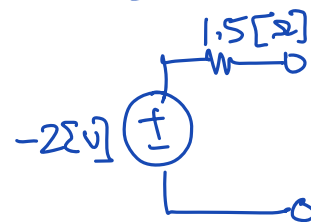
Use source transformations and series/parallel resistance combinations to simplify the circuit on the left into the equivalent circuit on the right to find v_{eq} and R_{eq} .



$$\frac{4}{6} [A] \cdot 3 [\Omega] = 2 [V]$$



$$-\frac{4}{3} [A] \cdot \frac{3}{2} [\Omega] = -2 [V]$$



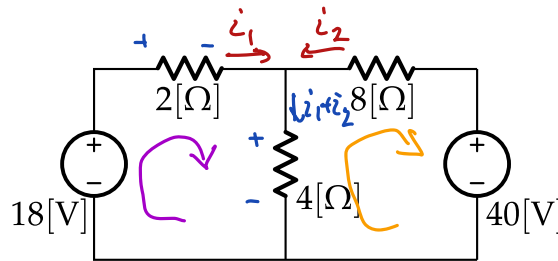
$$v_{eq} = \underline{-2 [V]}$$

$$R_{eq} = \underline{1.5 [\Omega]}$$

Problem 3

For the circuit below

1. Label the unknown currents on the circuit diagram
2. Write the raw KVL equations for the two loops in the space provided
3. Simplify the equations by collecting terms and write them into the space provided
4. Solve for the missing currents in each resistor
5. Determine the power supplied by each of the sources
6. Verify that the power supplied is equal to the power received by the resistors



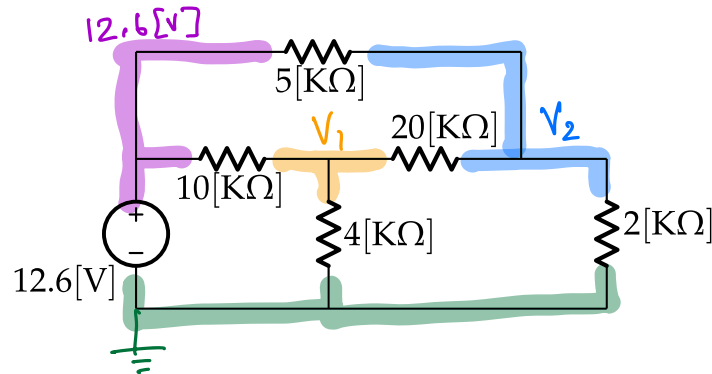
$$\begin{aligned} \div 2[\Omega] \left[-18[V] + i_1 \cdot 2[\Omega] + (i_1 + i_2)4[\Omega] = 0 \right] & \quad \div 4[\Omega] \left[-(i_1 + i_2)4[\Omega] - i_2 \cdot 8[\Omega] + 40[V] = 0 \right] \\ 3i_1 + 2i_2 = 9[A] & \quad -i_1 - i_2 - 2i_2 = -10[A] \\ 3(10[A] - 3i_2) + 2i_2 = 9[A] & \quad i_1 + 3i_2 = 10[A] \\ 30[A] - 9i_2 + 2i_2 = 9[A] & \\ -7i_2 = -21[A] & \\ i_2 = 3[A] & \\ i_1 + 3(3[A]) = 10[A] & \\ i_1 = 1[A] & \end{aligned}$$

Raw equations	$-18[V] + i_1 \cdot 2[\Omega] + (i_1 + i_2)4[\Omega] = 0$ $-(i_1 + i_2)4[\Omega] - i_2 \cdot 8[\Omega] + 40[V] = 0$
Simplified equations	$3i_1 + 2i_2 = 9[A]$ $i_1 + 3i_2 = 10[A]$
Currents	$i_1 = 1[A]$ $i_2 = 3[A]$
p (sources)	$18[V] \cdot 1[A] + 40[V] \cdot 3[A] = 138[W]$
p (resistors)	$(1[A])^2 \cdot 2[\Omega] + (3[A])^2 \cdot 8[\Omega] + (4[A])^2 \cdot 4[\Omega] = 138[W]$ yes, they are equal

Problem 4

Using nodal analysis (i.e., node voltage method),

- Label nodes needed to solve this problem as V_1 , V_2 , and GND, and write the two raw node voltage equations in the space provided
- Simplify the two equations and write them in the space provided
- Determine the values of V_1 and V_2
- How much power is dissipated in the 10 [K Ω] resistor?



$$\times 20[\text{K}\Omega] \left[\frac{V_1 - 12.6[\text{V}]}{10[\text{K}\Omega]} + \frac{V_1}{4[\text{K}\Omega]} + \frac{V_1 - V_2}{20[\text{K}\Omega]} = 0 \right]$$

$$2V_1 - 25.2[\text{V}] + 5V_1 + V_1 - V_2 = 0$$

$$8V_1 - V_2 = 25.2[\text{V}]$$

$$8(15V_2 - 50.4[\text{V}]) - V_2 = 25.2[\text{V}]$$

$$120V_2 - 403.2[\text{V}] - V_2 = 25.2[\text{V}]$$

$$119V_2 = 428.4[\text{V}] \Rightarrow V_2 = 3.6[\text{V}]$$

$$\left[\frac{V_2 - V_1}{20[\text{K}\Omega]} + \frac{V_2 - 12.6[\text{V}]}{5[\text{K}\Omega]} + \frac{V_2}{2[\text{K}\Omega]} = 0 \right] \times 20[\text{K}\Omega]$$

$$V_2 - V_1 + 4V_2 - 50.4[\text{V}] + 10V_2 = 0$$

$$-V_1 + 15V_2 = 50.4[\text{V}]$$

$$V_1 = 15V_2 - 50.4[\text{V}]$$

$$= 15(3.6[\text{V}]) - 50.4[\text{V}] = 3.6[\text{V}]$$

Raw equations	$\frac{V_1 - 12.6[\text{V}]}{10[\text{K}\Omega]} + \frac{V_1}{4[\text{K}\Omega]} + \frac{V_1 - V_2}{20[\text{K}\Omega]} = 0$ $\frac{V_2 - V_1}{20[\text{K}\Omega]} + \frac{V_2 - 12.6[\text{V}]}{5[\text{K}\Omega]} + \frac{V_2}{2[\text{K}\Omega]} = 0$
Simplified equations	$8V_1 - V_2 = 25.2[\text{V}]$ $-V_1 + 15V_2 = 50.4[\text{V}]$
V_1	$3.6[\text{V}]$
V_2	$3.6[\text{V}]$
p for 10 [K Ω] resistor	$p = \frac{V^2}{R} = \frac{(12.6[\text{V}] - 3.6[\text{V}])^2}{10[\text{K}\Omega]} = \frac{81}{10,000} = 8.1[\text{mW}]$

